LaunchPilot AI

An AI-powered guide for first-time founders

1. Choose a Problem Worth Solving

Three problems I considered

LaunchPilot AI — Helping first-time founders get started. The idea came from a conversation with a friend who wanted to start a company in the US while on an F-1 student visa. She knew what she wanted to build but was confused about company ownership, working in the business, and registration requirements. I realized that even someone with a clear business idea could struggle to understand where to begin.

CaseTrial — Organizing legal evidence. A relative who works as a lawyer introduced me to the document-heavy nature of legal cases. I considered creating a tool that connects emails, statements, and filings into a source-linked timeline. However, I would need more exposure to actual legal workflows before pursuing it.

ReguLens — Simplifying business compliance. While researching government resources, I noticed how difficult it can be to determine which regulations apply to a particular business. I considered a tool that translates compliance information into actionable steps, but the problem would need validation with business owners.

Why did I choose LaunchPilot?

LaunchPilot stood out because I had seen the problem firsthand. My friend had an idea and the motivation to start, but she didn't know what to do next. I wanted to make those first steps less intimidating, especially for someone without a legal or business background.

What I observed was one person's confusion and the complexity of navigating government information. I'm assuming other founders face similar challenges, but I would need to validate that through more conversations and user testing. One experience alone doesn't prove that people would use or pay for the product.

2. Develop My Approach

Today, founders typically rely on government websites, business formation services, online articles, videos, or professional advisors. Although information is available, I found that unfamiliar terminology and too many navigation options can make the process overwhelming.

My goal with LaunchPilot is simple: a founder should leave knowing **what to do next, why it matters, and what to prepare**, with the ability to return later and continue.

The proposed experience

I designed LaunchPilot around two starting points.

Ready to Launch: For someone who already has a business concept or developed product but needs help understanding business structures, registration, and compliance. They enter their details and receive a personalized roadmap based on their industry, location, and registration status.

Exploring an Idea: For someone who isn't sure how or where to start. The AI asks relevant questions, compares business models, discusses location considerations, and helps the founder explore alternatives. They can continue exploring or move into the launch process when ready.

For the smallest useful version, I focused on a founder starting an Indian Grocery store business in Phoenix, Arizona. After entering and confirming their business details, the founder receives a roadmap, opens a task, reviews its requirements, and can prepare and upload relevant documents.

Why a guided workflow instead of a checklist?

A static checklist would be simpler and more predictable. However, business type, location, and existing registration status can change which steps are relevant. I chose a guided workflow to account for those differences.

AI handles conversational discovery and explanations, while conventional software manages input validation, authentication, saved progress, and task rules. Users also review AI-extracted details before saving them.

3. Build a Working Prototype

Core workflow

Sign In → Select Ready to Launch → Enter Business Details → View Roadmap → Open Task Guide → Prepare and Upload Document → Mark Task Complete

Example input

"I want to start a consulting business in Richmond, Virginia. I haven't registered it yet or chosen a legal structure."

Expected output

LaunchPilot saves the business profile and generates a roadmap based on the provided information. Selecting a task opens a focused guide containing:

1. **What is this step?** A short, beginner-friendly explanation.
2. **Terms to know:** Important terminology explained simply.
3. **Documents needed:** Information or files the founder may need to prepare.
4. **Helpful resources:** Official websites and relevant YouTube tutorials.

A contextual AI assistant appears on the right, allowing founders to ask questions about the selected task. They can also download preparation worksheets and upload

completed documents. Uploading a file does not automatically mark the task complete.

Handling difficult cases

I considered incomplete user input and external API failures. For example, if someone leaves their operating city blank, the application should request that information rather than guess and potentially provide incorrect requirements. If an AI request fails, the user should see an understandable error and have the option to retry.

Architecture and implementation

Architecture - Link

Component	Technology
FrontEnd	React, TypeScript
BackEnd	Python,FastApi
Database	SqlAlchemy,SqlIte
AI Integration	LLM API
Government Research	Brave/Tavily

LaunchPilot uses React and TypeScript for the frontend, FastAPI for backend APIs and validation, and SQLAlchemy with SQLite for storing business profiles, conversations, documents, and roadmap progress.

An LLM API supports business discovery and contextual explanations. The government research service uses Brave or Tavily to locate relevant official sources and retrieve supporting information.

Roadmap rules combine saved business details with shared task definitions. Uploaded document contents are not sent to the LLM.

The prototype includes authentication, two onboarding journeys, personalized roadmaps, task-specific AI assistance, research, worksheets, uploads, and progress tracking. It does not automatically submit government applications, draft legal contracts, or determine immigration work authorization. Regulatory research coverage remains limited.

Technical Decisions and Trade-offs

While building LaunchPilot, I made two important technical decisions:

1. **LLM API vs. Training My Own Model:** I chose an existing LLM API because it allowed me to build and test the product faster without collecting training data or managing model infrastructure. The trade-off is dependency on an external provider, API costs, and privacy considerations.

2. **Official-Source Retrieval vs. Model Memory:** Instead of relying entirely on an LLM's knowledge for legal and compliance questions, I integrated government-source research to provide more traceable information. This improves transparency but introduces additional latency and doesn't guarantee complete regulatory coverage.

A design decision that changed the product

One issue I noticed during development was that the task guide had become too crowded. It contained planning tips, recommendation cards, and several informational sections.

That was the opposite of what I wanted LaunchPilot to achieve.

I simplified the page to four essential sections with an AI assistant on the right, so founders can focus on one action at a time instead of navigating unnecessary information.

Prototype and source code

- **Live application:** Link
- **GitHub repository:** https://github.com/shalini0408/LaunchPilot

The repository includes a README with setup instructions and configuration details.

How I used AI and validated the work

I used ChatGPT, Claude, and Cursor for coding assistance, debugging, understanding implementation issues, and improving the interface. I reviewed AI-generated code rather than accepting it automatically.

Earlier iterations were checked through frontend and backend tests, manual reviews, and browser tests with mocked external services. These checks helped catch implementation issues but do not establish the accuracy of every AI response or regulatory interpretation.

What I'd do next

I would test LaunchPilot with first-time founders and observe whether they can understand the guidance and complete a task without additional help. I'd also compare their experience against a simple checklist and have a qualified expert review a narrow set of regulatory requirements.

I would also verify reliable data storage and recovery in the hosted application.

Ultimately, I want LaunchPilot to help someone move from *"I have a business idea, but I don't know where to begin"* to *"I understand my next step, and I know how to take it."*